

[603]

SARDAR PATEL UNIVERSITY**M. Sc. Integrated Biotechnology, Seventh Semester Examination****Thursday, 7th January,
2021****10:00 a.m. to 12:00 p.m.****Environmental Chemistry : PS07EIBE23****Total Marks : 70****Note :** (i) All questions are to be attempted. (ii) Figures to the right indicate marks.**Q.1 Choose the correct option for the following:****[08]**

- (a) (i) A major burden of air pollutant is carried by
(a) Troposphere (b) Ionosphere (c) Stratosphere (d) Thermosphere
- (ii) Main component responsible for acid rain is
(a) NO_x (b) SO_x (c) CO (d) Both 'a' & 'b'
- (iii) Which is chlorinated hydrocarbon insecticide ?
(a) DDT (b) Acetic acid (c) TNT (d) All of these.
- (iv) The ability of solutes in water to neutralize added strong acid is called.....
(a) Acidity (b) Alkalinity (c) Neutrality (d) All of these.
- (v) From the following process is known as nitrate immobilization.
(a) Dissimilatory nitrate reduction (b) Ammonia assimilation
(c) Denitrification (d) Assimilator Nitrate reduction
- (vi) is NOT a correct general phosphorus transformation process.
(a) Precipitation – Assimilation (b) Sorption – Desorption
(c) Immobilization – Mineralization (d) None of these
- (vii) Which horizon of soil profile is lacking organic matter ?
(a) The 'O' horizon (b) The 'A' horizon (c) The 'C' horizon (d) None of these
- (viii) How are PAHs related to soil pollution?
(a) They are fertilizers waste (b) They are inorganic waste from the industries
(c) They are carcinogenic (d) They are harmful metals that pollute soil

Q. 1 State whether the following statements are true or false.**[16]**

- (b) (i) The most significant feature of atmospheric chemistry is the occurrence of photochemical reactions.
- (ii) A significant natural source of radio nuclides in the atmosphere is Xenone.
- (iii) The two major sources of atmospheric carbon dioxide are burning of fossil fuels and deforestation.
- (iv) The bottom layer of the atmosphere is the troposphere.
- (v) A highly toxic inorganic water pollutant that is widely used in certain mineral processing operations is heavy metals.
- (vi) Biochemical oxygen demand is a measurement of the degree of oxygen consumption by microbially mediated oxidation of contaminants in water.
- (vii) The advantage of LAS surface active agent over ABS is that the former is non biodegradable.
- (viii) Radiation damages living organisms by initiating harmful chemical reactions in tissues.
- (ix) Utilization of fossil fuels and Deforestation activities has increased the amount of fixed organic carbon reservoirs.

[1]

[P.T.O.]

- (x) Methylophs can use methane monoxygenase enzyme to oxidize ammonium.
- (xi) Burning of fossil fuel and mining practices have caused a disturbance in global sulfur reservoirs.
- (xii) The effect of organic pollutants in the atmosphere is due to the formation of Primary pollutants.
- (xiii) Calcareous soils have high buffering capacity due to presence of free NH_4^+ ions.
- (xiv) Gullyerosion type of soil erosion creates hazardous conditions to operate farm machinery.
- (xv) Primary microplastics are tiny particles designed for commercial use and secondary microplastics are particles that result from the breakdown of larger plastic items.
- (xvi) Industrial waste and garbage pollution are natural sources of soil pollution.

Q.2 Answer the following : (Attempt any seven)

- (i) Define green house effect. Enlist various green house gases and their sources. [14]
- (ii) What is meant by Hardness of water? How can be it calculated?
- (iii) Draw labeled hydrological cycle showing each step of cycle.
- (iv) Write Gaseous exchange reaction occur in water.
- (v) How methane oxidation solves pollution problem? Explain it with giving suitable example.
- (vi) What are the adverse effects of excess nitrification processes in environment?
- (vii) Differentiate between denitrification and anammox.
- (viii) Explain possible impacts of soil acidification.
- (ix) What are the three categories of in-situ physical treatment processes of contaminated soil?

Q.3 Write the main composition of atmosphere segments and main chemical reactions occur in atmosphere. [08]

OR

Q.3 Define and enlist various gaseous pollutants. Write the major sources and chemical reaction of any two pollutants in detail. [08]

Q.4 List out various organic water pollutant. Discuss in brief about pesticides pollutant. [08]

OR

Q.4 Discuss about toxic metals pollutant in water bodies, their sources and adverse effect of each metal. [08]

Q.5 (i) Write a note on organic polymers. [04]
(ii) Describe process of ammonia assimilation and ammonification of nitrogen cycle. [04]

OR

Q.5 (i) Explain biological nitrate reduction processes in brief. [04]
(ii) Discuss assimilatory sulfate reduction and sulfur mineralization. [04]

Q.6 (i) Explain biological remediation processes of acidic soil. [04]
(ii) Explain in-situ chemical oxidation and chemical reduction processes used to treat contaminated soil. [04]

OR

Q.6 (i) Giving appropriate examples write a note on importance of soil reactions. [04]
(ii) Write a note on physical properties of the soil. [04]

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